

HMMT 2013
Saturday 16 February 2013

HMIC

1. [5] Let $\mathcal{S}$ be a set of size n , and k be a positive integer. For each $1 \leq i \leq kn$, there is a subset $S_i \subset \mathcal{S}$ such that $|S_i| = 2$. Furthermore, for each $e \in \mathcal{S}$, there are exactly $2k$ values of i such that $e \in S_i$. Show that it is possible to choose one element from S_i for each $1 \leq i \leq kn$ such that every element of $\mathcal{S}$ is chosen exactly k times.

2. [7] Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all real numbers x, y ,

$$(x - y)(f(x) - f(y)) = f(x - f(y))f(f(x) - y).$$

3. [8] Triangle ABC is inscribed in a circle ω such that $\angle A = 60^\circ$ and $\angle B = 75^\circ$. Let the bisector of angle A meet BC and ω at E and D , respectively. Let the reflections of A across D and C be D' and C' , respectively. If the tangent to ω at A meets line BC at P , and the circumcircle of APD' meets line AC at $F \neq A$, prove that the circumcircle of $C'FE$ is tangent to BC at E .

4. [10] A subset $U \subset \mathbb{R}$ is *open* if for any $x \in U$, there exist real numbers a, b such that $x \in (a, b) \subset U$. Suppose $S \subset \mathbb{R}$ has the property that any open set intersecting $(0, 1)$ also intersects S . Let T be a countable collection of open sets containing S . Prove that the intersection of all of the sets of T is not a countable subset of $\mathbb{R}$.

(A set Γ is *countable* if there exists a bijective function $f : \Gamma \rightarrow \mathbb{Z}$.)

5. [12]

- (a) Given a finite set X of points in the plane, let $f_X(n)$ be the largest possible area of a polygon with at most n vertices, all of which are points of X . Prove that if m, n are integers with $m \geq n > 2$, then $f_X(m) + f_X(n) \geq f_X(m+1) + f_X(n-1)$.

- (b) Let P_0 be a 1-by-2 rectangle (including its interior), and inductively define the polygon P_i to be the result of folding P_{i-1} over some line that cuts P_{i-1} into two **connected** parts. The *diameter* of a polygon P_i is the maximum distance between two points of P_i . Determine the smallest possible diameter of P_{2013} .

(In other words, given a polygon P_{i-1} , a fold of P_{i-1} consists of a line l dividing P_{i-1} into two connected parts A and B , and the folded polygon $P_i = A \cup B_l$, where B_l is the reflection of B over the line l .)